# Noise Insulation and Student Achievement in Airport-Adjacent Schools: Pre-Analysis Plan

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## 1 Introduction

According to the U.S. Bureau of Transportation Statistics, seasonally-adjusted air passenger miles in the United States increased over 71% between 2002 and 2020. The rapid growth of commercial aviation has strained communities located near major airports. Despite technological improvements reducing the noise produced by jets taking off and landing, airports still generate large externalities in the form of air pollution, noise pollution, increased traffic, and lower property values (Ryley, 2014; Zheng, Peng and Hu, 2020; Cattaneo et al., 2023).

Seattle-Tacoma International Airport (Sea-Tac) is a prime example of this phenomenon. Located approximately 23km south of downtown Seattle, the airport is set along the busy Interstate-5 corridor and is immediately surrounded by the cities of Burien, Des Moines, Normandy Park, and SeaTac.

A specific concern among many community members is the effect of aircraft noise on schools. Anecdotally, this noise has complicated learning and teaching around Sea-Tac Airport for decades:

> A jumbo jet wooshes over Pacific Middle School, but that doesn't stop Amy Driscoll's lecture to her seventh-grade health class.
>
> "In this book... In this book... You guys aren't listening!" she says, elevating her voice with each word
>
> - The Seattle Times: December 16, 1998

To address concerns about further expansions to Sea-Tac Airport, in 2002 Highline Public Schools (HPS) reached a memorandum of understanding (MOA) with three public agencies - The Port of Seattle, the Federal Aviation Administration, and the State of Washington. Under the terms set in the MOA, those three entities would provide $50 million each to HPS for noise mitigation improvements. At the time of the agreement, 15 schools were selected for funding. To date, 10 have been completed.

The objective of this analysis is to evaluate the effect of noise mitigation funding on student achievement. Specifically, I will use the staggered rollout of HPS noise mitigation projects since 2004 to identify the effect of a school receiving these funds on student test scores, high school GPA, advanced course enrollment, on-time graduation, and absence and discipline history.

Previous research suggests that students bear a cost from exposure to noise pollution. A meta-analysis by Thompson et al. (2022) finds that reading and language abilities are negatively effected by environmental noise. More specific to airport noise pollution, Klatte et al. (2017) study the effects of aircraft noise in second-graders around Frankfurt/Main Airport in Germany. They find that increasing noise exposure is associated with decreased reading performance, despite a relatively low level of ambient noise pollution present in the schools. Furthermore, a meta-analysis by Mealings (2022) demonstrates that cognition is affected by both acute and chronic noise exposure. To the best of my knowledge, this will be the first study analyzing the effect of noise insulation technology on cognitive performance in a pollution-exposed population.

# 2 Data

The primary data used for this analysis is obtained from the Washington State Education Research & Data Center's (ERDC's) Preschool-to-Grade-20-to-Workforce (P20W) data system.[1] This data set is a student-level panel containing information on standardized test scores, demographics, high school GPAs, course schedules, and absence/discipline history for any student who attended K-12 in Washington between 2005-2020.

Data on the frequency and altitude of flights near schools comes from the Healthy Air, Healthy Schools project (Austin et al., 2021). Information on school noise mitigation projects was obtained via public records requests from Highline Public Schools.

## 2.1 School Noise Mitigation Projects

Highline Public Schools (HPS) is a large, K-12 school district serving around 18,000 students grades K-12 in the communities of Burien, Des Moines, Normandy Park, SeaTac and White Center in Washington State. The district's location, centered around Seattle-Tacoma International Airport, has exposed its students to the negative externalities of the growing airport for decades. In 2002, HPS reached a memorandum of understanding with the Port of Seattle, Federal Aviation Administration, and the State of Washington, each of whom agreed to provide up to $50 million each in funding to support noise mitigation for a select group of schools.

The schools eligible for funding were those which fell within the airport's DNL65 noise contour as defined by Sea-Tac Airport's 1991 Part 150 study. DNL, or day-night average sound level, is a formula that attempts to capture the total level of noise-induced annoyance experienced in a given location.

Figure 1 shows how the timeline of school construction projects in HPS overlaps with the availability of my outcomes data. I define a school as treated after the noise insulation project has been considered completed by the school district.

The difference-in-differerences analysis will include the following treated schools:

- Madrona Elementary
- North Hill Elementary
- Cedarhurst Elementary
- Midway Elementary
- Parkside Elementary

Mt. Rainier High is excluded because construction on the new school began in 2005, the same year as the first year test score data. For the two years that the school was under construction,

[1] This data has not been received by the author as of the date of this document.

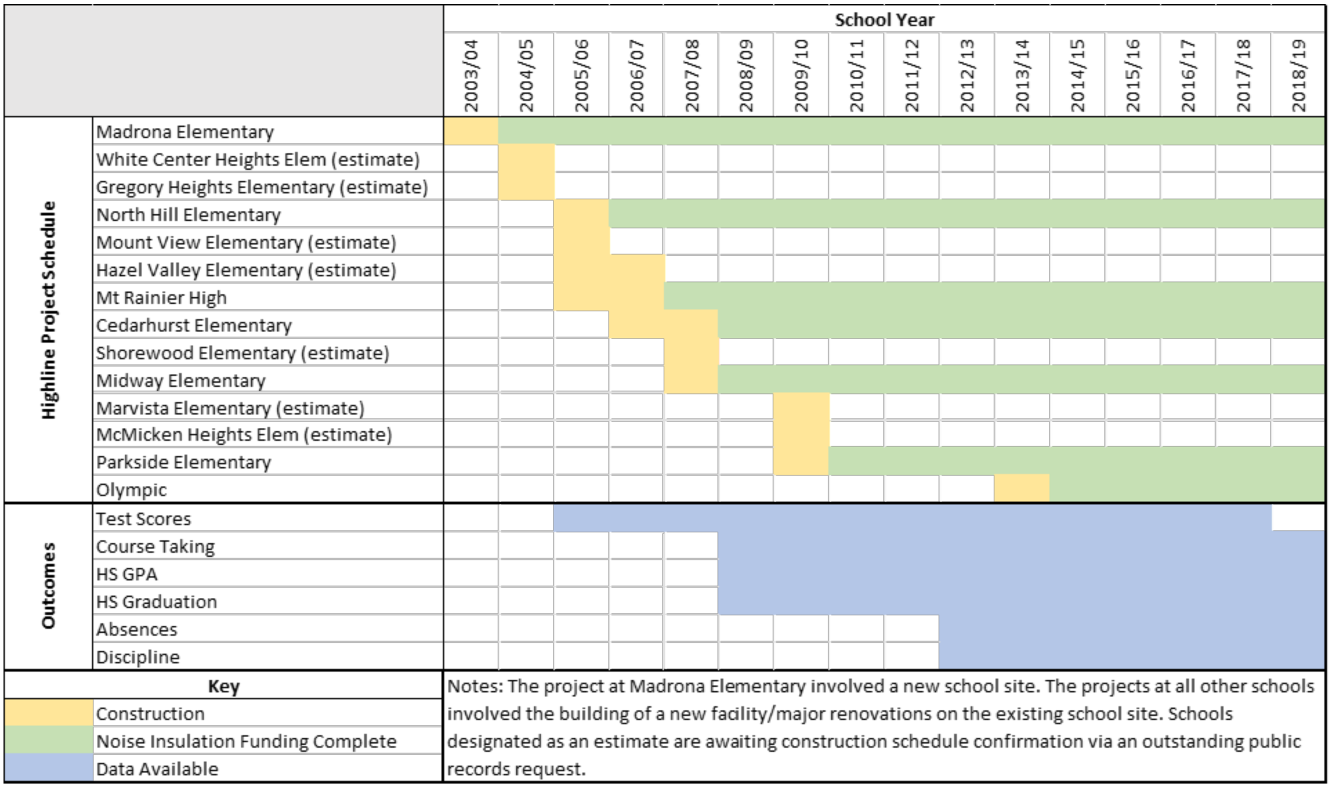


Figure 1: Timeline of Highline School District Construction and Student Outcome Availability

students were sent to the Olympic Interim Site, approximately 2.7km northwest of Mt. Rainier High. Since this interim location is closer to the airport, noise exposure is likely to be higher for these two pre-treatment years. This threatens the parallel trends assumption inherent to my analysis, since the opening of the new high school location also coincided with a move back away from the airport.

The Olympic site is excluded since it has served only as an interim school location since the closure of Olympic Elementary School after the 2004/05 school year. Other schools which were designated for noise insulation funding are not shown here if the project was not completed before the end of my study period.

## 2.2 Construction of the Outcome Variables

My primary short-run outcomes of interest are elementary mathematics and reading standardized test scores. As tests may have changed over the years, I will standardize test scores within subject, grade, and year to be mean zero and standard deviation of one (where the mean and standard deviation is defined by the entire data set). Should data be available, I will also include standardized test scores from middle and/or high school as an outcome.

Long-run outcomes of interest include high school GPA, advanced course enrollment, on-time graduation, and absence and discipline history. These are considered longer-run outcomes since data is only available on GPA, graduation, and course schedules starting in 2009, and on absence and discipline starting in 2012.

When measuring GPA, I will use the last observed high school GPA for a given student. Advanced course taking will be summarized with a dummy variable indicating whether a student took any advanced placement (AP) or International Baccalaureate (IB) courses in high school, as the literature suggests students see benefits from taking even a single advanced class (for example, see Conger et al. (2021)). On-time graduation indicates if the student graduated in the expected year, which is defined by the student's first high school enrollment record. I will construct an absence variable corresponding to the average number of days in a year that a student was absent from school (both excused and unexcused) while they were enrolled in high school. Discipline will be measured by the average number of days in a year that a student was removed from the classroom and/or the school due to disciplinary action during the course of the student's high school career.

# 3 Empirical Strategy

The primary goal of this paper is to identify the causal effect of noise insulation on student outcomes. A naive correlation may be plagued by severe endogeneity concerns and cannot be treated with a causal interpretation. For example, residential sorting may result in families with more resources moving away from noise-effected areas. As income is correlated with test scores (Reardon, 2014), this could bias the estimates. Another concern is reverse causality: if schools were selected for noise insulation due to being especially poor performers, this would downwardly bias our estimate for the treatment effect of noise insulation.

To obtain more credible estimate, I leverage the staggered rollout of noise insulation projects among select Highline schools beginning in 2004. The quasi-experimental variation generated by this staggered rollout allows for the estimation of the causal effect of noise insulation on school performance using a difference-in-differences strategy. This approach compares the difference in outcomes between treated and untreated schools before and after the completion of noise insulation projects.

## 3.1 Effects of Noise Mitigation Funding

For the purposes of this analysis, I limit my sample to students who attended elementary schools in King County which had at least one flight under 750m pass within a mile of the school in 2019.

For clarity, let $g$ denote the treatment group, or the time period when a school first becomes treated. I am interested in the effect of noise insulation treatment $D_{g,t}$ on an outcome $Y_{i,t}$ for individual $i$ who is attending a school in treatment group $g$ in year $t$.

To assess the effect of noise mitigation funding on test scores, I will estimate the following two-way fixed effect (TWFE) model:

$$Y_{i,t} = \gamma_g + \gamma_t + \beta D_{g,t} + \mathbf{X}_{i,t}\theta + u_{i,t} \tag{1}$$

$\gamma_g$ represents treatment group fixed effects.[2] $\gamma_t$ represents year fixed effects. $\mathbf{X}_{i,t}$ is a vector of individual-level controls, including race/ethnicity, gender, disability status, and an indicator for being an English-language learner.

Similarly, I will estimate the following model for high school outcomes:

$$Y_{i,t} = \gamma_g + \gamma_t + \beta D_{g,t} + \mathbf{X}_{i,t}\theta + \alpha_{hs,t} + e_{i,t} \tag{2}$$

[2] In this case, at most one school was treated in any given year, so treatment group fixed effects are equivalent to school fixed effects.

where $\alpha_{hs,t}$ represents high school-year fixed effects. This term is added to this model for two reasons. First, there is the issue that Mt. Rainier High School received noise mitigation funding of its own prior to the 2007/08 school year. If some students from treated elementary schools ended up attending Mt. Rainier High, then failure to account for this fact could result in misattributing treatment effects from high school to having attended a treated elementary school. Second, one outcome of interest is whether a student took any advanced classes while in high school. As the selection of advanced courses offered varies across schools and over time, a high school-year fixed effect will hold constant a student's ability to enroll in a given advanced course.

If, absent any noise insulation treatment, outcomes (e.g., test scores) for students in different treatment groups would have followed parallel trends, and if the average treatment effects are homogenous across treated schools and over time, then the coefficient $\beta$ identifies the average treatment effect on the treated (ATT) of noise insulation treatment on the outcome.

Developments in the difference-in-differences literature have demonstrated issues with the TWFE estimator in the presence of treatment effect heterogeneity and staggered treatment implementation. Specifically, Goodman-Bacon (2021) shows that a causal interpretation of TWFE estimates requires treatment effects that are constant over time. This assumption is unlikely to hold in this setting - the effect of noise insulation may increase in later years due to the airport's constant expansion over the study period. To address this concern, I will rely on the robust estimator introduced in Callaway and Sant'Anna (2021) as my primary specification. This estimator delivers consistent estimates even in the presence of multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. This estimator will also be used to aggregate group-time effects into an event study plot.

It is possible that students who switch between schools are systematically different from those who do not. Should the selection analysis described in section 3.3 show that noise mitigation funding is associated with significant nonstructural school movement, I will repeat the analysis described above with a sample limited only to students who did not switch elementary schools after first grade.[3]

### 3.1.1 Isolating The Effect of Noise Insulation

One issue with model (1) is that the noise insulation investment was made conditional upon the passing of a local school bond. In effect, noise insulation investments were paired with larger school capital expenditures. Analysis of historical satellite imagery confirms that all schools which received noise insulation funding also underwent major construction during the same period. A concern is that the parameter estimated in (1) is capturing the net effect of noise insulation and broader school facility improvements.

In this section, I will incorporate data from other school construction projects within Highline School District during the study period. Specifically, I will use a triple differences (DDD) setup to separately identify the effects of school renovation and having received funding for noise insulation.

I limit estimation of this model to a sample of only schools within the Highline School District to address concerns about the potential endogeneity of new school construction (Specifically, that communities with more resources are more likely to fund new schools and also have higher test scores). Such a restriction is useful for three reasons:

[3]Kindergarten attendance is not required in Washington.

1. School-level construction decisions are made at the district level and are contingent upon the passing of a district-wide school bond authorization.

2. A large portion of Highline schools were built during the postwar housing boom of the 1950s (Richardson, 2008). These schools were due for replacement due to their age, largely independent of other considerations.

3. The last Highline School District bond issue approved before 2002 was in 1986 (Roberts, 2001). Highline voters rejected bond requests in 1992 and again in 2000. Thus, there was little new construction in the decade before the 2002 bond, reducing concerns that some students were already benefiting from new school construction at the start of my study period.

The intuition for the DDD approach is as follows. I can estimate the average treatment effect on the treated (ATT) of a school receiving any type of renovations using a standard difference-in-differences approach - simply compare how much outcomes changed over time in renovated schools to how much they changed over the same period in otherwise similar non-renovated schools. I can then recover our parameter of interest, the difference in ATTs between the schools which were selected for noise insulation funding and those which were not (call this difference the DATT). The DATT captures the relative effect of receiving school renovations between the noise insulation group and the comparison group.

To complicate matters somewhat, recent work by Caron (2025) shows that, when treatment effects are correlated with subgroup status, the DATT does not represent the causal effect of belonging to the subgroup of interest relative to the comparison subgroup. In this context, I do expect treatment effects (of receiving school renovations) to be correlated with subgroup status (having been selected for noise insulation funding) - schools which were selected for noise insulation are located closer to the airport, and so are likely to benefit more from renovations of all kinds.[4] Hence, I want to estimate the causal DATT (CDATT), or the differences in outcomes between the noise insulation and comparison groups *caused by* receiving noise insulation funding.[5]

The key assumption necessary for identification of the CDATT is that there is no treatment effect heterogeneity between the subgroups after conditioning on control variables. In other words, a student attending a school designated for noise insulation should be no more sensitive to school renovations than a student attending a school not designated for noise insulation, after conditioning. To support this assumption, I will condition estimation of the CDATT on student race/ethnicity, gender, status as an English learner, disabled status, and the number of flights under 750m which passed within a mile of the school in 2019. This last control is especially important as it serves as a proxy for a school's sensitivity to school renovations due to being exposed to more low-flying planes. Hence, the CDATT identifies the causal effect of a school receiving noise insulation funding on outcomes, holding constant whether a school received other renovations and the general level of low-flying aircraft experienced by that school.

Estimation of the CDATT will follow Caron (2025).

[4]For example, replacing temporary or portable classrooms with permanent structures is likely to reduce the impact of noise pollution irrespective of any formal noise insulation process. In addition, sorting around noise-affected schools could mean that schools closer to the airport are more likely to be socioeconomically disadvantaged, which Biasi, Lafortune and Schönholzer (2025) show benefit disproportionally from capital investments.

[5]For more information on the CDATT, reference Caron (2025).

## 3.2 Subgroup Analysis

The next area of interest is whether treatment effects are heterogeneous across the student population. For instance, students with a disability or who are learning English as a second language may be disproportionally harmed by noise pollution in the classroom and thus could benefit more from insulation.

To analyze this, I again set up a triple differences (DDD) design following the method proposed by Caron (2025). In this case, the treatment indicator $D_{g,t}$ indicates if a school in treatment group $g$ had received noise insulation funding by year $t$. Subgroup indicators will be generated based on the following variables:

- Race/Ethnicity
- Gender
- Disability Flag
- English Learner Flag

## 3.3 Selection Analysis

It is important to consider how school investments may influence student movement across schools. Interviews with local stakeholders suggest that many families choose to move away from the cities immediately surrounding the airport due to increasing nuisance from noise and air pollution and a perceived decline in school quality.

There is a distinction in the education literature between two types of student mobility: structural and nonstructural (Maroulis et al., 2019). Structural mobility refers to normally-progressing students who are forced to change schools due to grades offered or a school closing. Nonstructural mobility captures students who enroll in a different school the following year even though their grade was offered at their original school. I begin this analysis by focusing on nonstructural mobility.

My primary outcomes of interest are as follows:

- $X_{i,s,t}$, which is equal to one if student $i$ makes a nonstructural exit out of school $s$ after year $t$.
- $N_{i,s,t}$, which is equal to one if student $i$ makes a nonstructural entrance into school $s$ after year $t$.

As a baseline specification, I will estimate the following linear probability models:

$$X_{i,s,t} = \alpha_0 + \alpha_1 Mitigated_{s,t} + \alpha_2 Renovated_{s,t} + \tau_s + \mathbf{X}_{i,t}\theta + \epsilon_{i,s,t} \tag{3}$$

$$N_{i,s,t} = \lambda_0 + \lambda_1 Mitigated_{s,t} + \lambda_2 Renovated_{s,t} + \tau_s + \mathbf{X}_{i,t}\psi + \mu_{i,s,t} \tag{4}$$

where $Mitigated_{s,t}$ is equal to one if a school has received funding for noise mitigation by year $t$, $Renovated_{s,t}$ is equal to one if the school has received any major renovations by year $t$, $\tau_s$ is a school fixed effect, and $\mathbf{X}_{i,t}$ is a vector of individual-level controls including race/ethnicity, gender, disability status, and an indicator for being an English-language learner. Due to availability of data on school renovation schedules, I restrict the sample for this analysis to schools within HPS. Standard errors will be clustered at the school level.

The parameters of interest, $\alpha_1$ and $\lambda_1$, identify the effect of a school receiving noise mitigation funding on the probability of a student in that school making a nonstructural exit or entrance, respectively, holding constant whether that school has received other renovations, individual-level characteristics, and a time-invariant school fixed effect.

I will also analyze treatment effect heterogeneity by introducing interaction terms between the individual-level controls in $\mathbf{X}_{i,t}$ and the indicator $Mitigated_{s,t}$.

It is possible that the relevant student movement takes the form of structural mobility - families may choose to make a school change when a student completes elementary or middle school to minimize any potential disruption. To analyze this effect, I will re-estimate equations (3) and (4) with the dependent variables changed to reflect whether the student left or entered the district after completing elementary school, respectively.

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